
Artificial Intelligence

Assignment 4

Total Points: 5

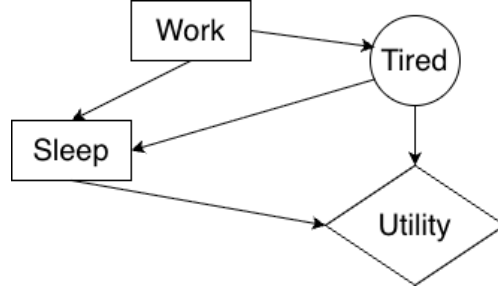
Uncertainty & Neural Nets

Due: 17:00, 8 May, 2026

Submission PDF to: <https://f.wps.cn/g/whW70KW9/>

1. Decision Networks (1.5 pts)

Give the optimal policy that achieves the highest expected utility of the following decision network:

**CPTs:**

Work	$P(\text{tired})$	Tired	Sleep	Utility
work	0.8	tired	sleep	30
rest	0.3	tired	no sleep	-10
		not tired	sleep	50
		not tired	no sleep	70

Solution: The node “Tired” is a parent node of decision node “Sleep”, so we do not eliminate this chance node. Instead, we need a policy to determine the choice of “Sleep” for each possible observation of “Tired”:

- If **Tired = tired**: Sleep = sleep gives utility 30, Sleep = no sleep gives utility -10. Optimal: Sleep = sleep (utility 30).
- If **Tired = not tired**: Sleep = sleep gives utility 50, Sleep = no sleep gives utility 70. Optimal: Sleep = no sleep (utility 70).

Thus the optimal policy for Sleep is:

$$\pi_{\text{Sleep}}^*(\text{tired}) = \text{sleep}, \quad \pi_{\text{Sleep}}^*(\text{not tired}) = \text{no sleep}.$$

Once the optimal policy of node “Sleep” is determined, we consider it is “eliminated”. Now, the chance node “Tired” will not be a parent of any undetermined decision node. So, we eliminate the variable “Tired”:

- If **Work = work**:

$$\begin{aligned} \text{Expected_Utility}(\text{work}) &= P(\text{tired} \mid \text{work}) \times 30 + P(\text{not tired} \mid \text{work}) \times 70 \\ &= 0.8 \times 30 + 0.2 \times 70 \\ &= 24 + 14 = 38. \end{aligned}$$

- If **Work = rest**:

$$\begin{aligned} \text{Expected_Utility}(\text{rest}) &= P(\text{tired} \mid \text{rest}) \times 30 + P(\text{not tired} \mid \text{rest}) \times 70 \\ &= 0.3 \times 30 + 0.7 \times 70 \\ &= 9 + 49 = 58. \end{aligned}$$

The optimal value for Work is rest. So the final optimal policy is:

1. Choose Work = rest.
2. After observing the resulting Tired state:
 - If Tired = tired, choose Sleep = sleep.
 - If Tired = not tired, choose Sleep = no sleep.

The maximum expected utility is 58.

2. Markov Decision Process (2 pts) A small robot must manage its energy level while working. At any time, the robot is in one of two **states**: High energy (H) and Low energy (L). In each state, the robot can choose one of two **actions**: Work (W) or Rest (R). The environment is stochastic. The **transition probabilities** $P(s' | s, a)$ are given below.

s	a	$P(s' = H s, a)$	$P(s' = L s, a)$
H	W	0.3	0.7
H	R	0.9	0.1
L	W	0.6	0.4
L	R	0.4	0.6

The immediate **reward function** $R(s, a)$ is provided by the table below.

s	a	Reward
H	W	+5
H	R	+1
L	W	+2
L	R	+3

Let the **discount factor** be $\gamma = 0.9$. Perform value iteration algorithm for **one iteration** starting from $V_0(H) = 0$ and $V_0(L) = 0$ and give the updated values $V_1(H), V_1(L)$ for the two states.

Solution: Based on the Bellman optimal value equation: $V^*(s) = \max_{a \in A} \sum_{s'} P_{s \rightarrow s'}^a (R_{s \rightarrow s'}^a + \gamma V^*(s'))$. Value iteration will use an iterative method to get V^* , starting with $V_0(s) = 0$ and update the value by $V_1(s) \leftarrow \max_{a \in A} \sum_{s'} P_{s \rightarrow s'}^a (R_{s \rightarrow s'}^a + \gamma V_0^*(s'))$.

- For $V_1(H)$:
 - If action $a = W$, the expected value is $P(s' = H | s = H, a = W) \times (\text{Reward}(H, W) + \gamma V_0(H)) + P(s' = L | s = H, a = W) \times (\text{Reward}(H, W) + \gamma V_0(L))$. By substituting the values from the given tables, the expected value is $0.3 \times 5 + 0.7 \times 5 = 5$.
 - Similarly, if action $a = R$, the expected value is $0.9 \times 1 + 0.1 \times 1 = 1$.

Thus, $V_1(H) = \max(5, 1) = 5$.

- For $V_1(L)$:
 - If action $a = W$, the expected value is $0.6 \times 2 + 0.4 \times 2 = 2$.
 - Similarly, if action $a = R$, the expected value is $0.4 \times 3 + 0.6 \times 3 = 3$.

Thus, $V_1(L) = \max(2, 3) = 3$.

3. Neural Networks (1.5 pts) Consider a neural network to prediction sleep quality:

- Input layer: x_1, x_2 (Let x_1 be the average daily screen time, x_2 be daily physical activity)
- Hidden layer: 2 neurons with ReLU activation
- Output layer: 1 neuron with sigmoid activation, output 1 to denote good sleep quality, and 0 to denote bad sleep quality.

Weights of the hidden layer and the output layer (given):

$$W^{(h)} = \begin{bmatrix} 0.3 & -0.5 \\ 0.8 & 0.2 \end{bmatrix}, \quad b^{(h)} = \begin{bmatrix} 0.1 \\ -0.2 \end{bmatrix}, \quad W^{(out)} = \begin{bmatrix} 0.6 & -0.4 \end{bmatrix}, \quad b^{(out)} = 0.3$$

Consider the input sample: $x = [x_1, x_2]^T = [1.0, 0.5]^T$ with true label $y = 1$:

1. Compute hidden layer activations: $z = \text{ReLU}(W^{(h)}x + b^{(h)})$.
2. Compute output: $\hat{y}$.
3. Compute binary cross-entropy loss: $L = -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})]$.

Solution:

1.

$$z = \text{ReLU}\left(\begin{bmatrix} 0.3 & -0.5 \\ 0.8 & 0.2 \end{bmatrix} \times \begin{bmatrix} 1.0 \\ 0.5 \end{bmatrix} + \begin{bmatrix} 0.1 \\ -0.2 \end{bmatrix}\right) = \text{ReLU}\left(\begin{bmatrix} 0.15 \\ 0.7 \end{bmatrix}\right) = \begin{bmatrix} 0.15 \\ 0.7 \end{bmatrix}$$

2.

$$\begin{aligned} \hat{y} &= \sigma(W^{(out)}z + b^{out}) \\ \hat{y} &= \sigma\left(\begin{bmatrix} 0.6 & -0.4 \end{bmatrix} \begin{bmatrix} 0.15 \\ 0.7 \end{bmatrix} + 0.3\right) = \sigma(0.11) \approx 0.53 \end{aligned}$$

3.

$$L = -\log 0.53 \approx 0.64$$